

The Jingzhang Diagram

Centennial Jingzhang AI Innovation Belt Urban Design Proposal (Final Evolution v1.5)

A city that runs by the diagram - Spatial Operating System & Four-System Hardwired Governance

Author: Juwan-Hwang • Formal AI urban design package

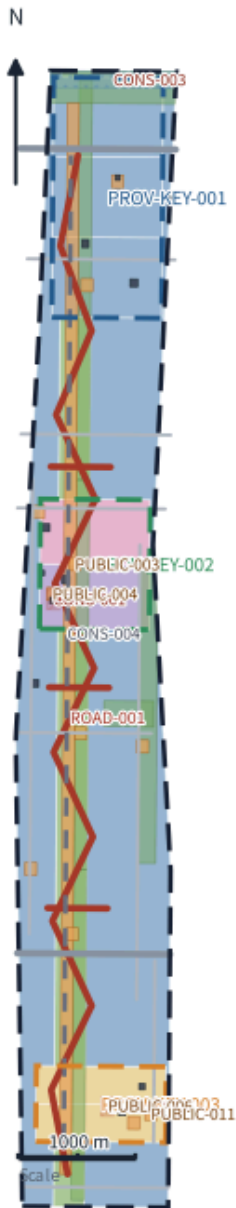
SEB v0.5.0 tabletop self-check complete (participant-declared, no third-party certification)

Provisional process boundary (EPSG:4548); not an official planning approval redline

OVERALL SPATIAL STRUCTURE & LOCATION PLAN • THE JINGZHANG DIAGRAM (provisional)

PROVISIONAL BOUNDARY (NON-STATUTORY)

Base: this package's geometry/*.geojson nine layers • EPSG:4548 • north arrow, scale bar, legend, feature_id labels



Legend (layers & feature_id)

- Site boundary SITE-001 (provisional)
- Key areas PROV-KEY-001/002/003
- Land use LU-001A~LU-005 (colored by code)
- Green space GREEN-001~006
- Public space PUBLIC-000~012
- Building footprints BLDG-001~010
- Road centerlines ROAD-001~014 (greenway/arterial/expressway)
- Constraints CONS-001~004 (heritage/blue line/rail, indicative)

Drawn from the organizer's provisional geometry; not an official redline. Areas and alignments follow metrics.json recomputation and the GeoJSON layers; redraw in full once official data is released.

THE JINGZHANG DIAGRAM: OVERALL SPATIAL OPERATING SYSTEM

PROVISIONAL BOUNDARY (NON-STATUTORY)

Four Epochs of Innovation along the Corridor & Three Key Areas of the Spatial OS

I. FOUR INNOVATION EPOCHS

1909 Jingzhang Railway

First trunk railway independently surveyed, designed and built by China; switchback topology beats the grade
[National Railway Museum Records]

1980s Zhongguancun

Historical narrative milestone of Chinese tech entrepreneurship and institutional innovation
[Historical Narrative Milestone]

2019 Jingzhang High-Speed Railway

World's first 350 km/h intelligent high-speed railway (full qualifier); intelligent running & Beidou at 350 km/h
[Ministry of Transport Official Statement]

2026 Centennial AI Innovation Belt

Proposal design proposition: human-AI co-ascent; diagram governance & The Human Switch safety
[Design Concept Proposition]

II. SPATIAL OPERATING SYSTEM - THREE KEY AREAS

Zhongzhiyuan AI Testbed

192.1 ha • Microgrid compute
Autonomous loop test track

Provisional boundary polygon • EPSG:4548 model internal only

Beijing AI Origin Community

104.3 ha • Staffed dispatch hall
Living district & night school

Provisional boundary polygon • EPSG:4548 model internal only

Dazhongsi Marshalling Yard

72.0 ha • Commercialisation
Two-part handover desk

Provisional boundary polygon • EPSG:4548 model internal only

Four industry test & validation scenarios:

Autonomous safety corridor

Robot city-service yard

LLM governance sandbox

Heritage digital twin

III. DIAGRAM GOVERNANCE: 4 HARDWIRED INTERLOCK SYSTEMS

信号制 Signal

Multi-tiered dynamic permissions
Charter 10 human-first principles

闭塞制 Block

Spatial & compute separation
Charter 10 human-first principles

联锁制 Interlock

SEB v0.5.0 tabletop self-check 7/7 (participant-declared)
Charter 10 human-first principles

路票制 Token

Human Switch tiered interface & opt-out
Charter 10 human-first principles

submissions/Juwan-Hwang/jingzhang-diagram • geometry/*.geojson • metrics.json • 图表为过程性研究表达，非法定依据

LAND USE STRUCTURE & ZONING PARTITION (TOPOLOGY SEAMLESS)

PROVISIONAL GEOMETRY • EPSG:4548

11.41 km² Overall Design Scope • Green Ratio 15.68% • Public Space Ratio 9.31% • 9 polygons seamless tiling

1. LAND-USE FUNCTIONAL MIX (recomputed from land_use.geojson)

0802 Research & two-wing tech services
808.59 ha • ≈70.8%
LU-001A/B/C + LU-005 (Zhongzhiyuan test track & wing interfaces)
1401 Park green spine
156.33 ha • ≈13.7%
LU-004 (Jingzhang heritage park blue-green open space)
05 Commercial services
72.05 ha • ≈6.3%
LU-003A/B (Dazhongsi AI-native market & station services)
0803 Culture
52.16 ha • ≈4.6%
LU-002A (Origin cultural core: hall / plaza / dispatch)
0702 Community services
52.16 ha • ≈4.6%
LU-002B (incubation & scholar-alleys mixed)
Slow-mobility network (centerlines)
≈9.5 km spine • ≈11.0 km total
ROAD-001 + stitch corridors ROAD-012/013/014 (not land-use polygons)

面积基于 land_use.geojson 各图斑 area_sqm_declared (EPSG:4548) ; 慢行里程为 roads.geojson greenway 中心线复算值 [metric:slow_corridor_length_m]

2. SPATIAL METRIC RECALCULATION

Total Site Area	11,412,825.39 m² (11.41 km²)
Green Space Union Area	1,789,421.32 m² (178.94 ha)
Green Ratio	0.15679 (15.68%)
Public Space Union Area	1,062,301.12 m² (106.23 ha)
Public Space Ratio	0.09308 (9.31%)
Slow corridor length	11,005.9 m (recomputed)
Key Areas Total	368.40 ha (3 Key Areas)
Floor Area Ratio (FAR)	unknown (pending statutory plan)
Topology Integrity	PASS (0 gaps / 0 overlaps)

Design judgment: innovation functions dominate; life & public services form the ≈29.2% base; the two-wing service land holds the largest share and its public-service mix must be subdivided at control-plan stage.

THREE KEY DETAILED-DESIGN AREAS (PROVISIONAL BOUNDARIES)

PROVISIONAL GEOMETRY • EPSG:4548

Zhongzhiyuan 192.1 ha • AI Origin Community 104.3 ha • Dazhongsi 72.0 ha • Total 368.4 ha

Zhongzhiyuan AI Testbed (192.1 ha)

ZHONGZHIYUAN AI ACCELERATION AREA

Positioning	Autonomous mobility & low-altitude testbed
Facilities	Microgrid compute hub, closed test track
Governance	L2 closed paired testing, power monitoring
Boundary	Provisional polygon awaiting statutory redline

TECHNICAL DISCLOSURE • PROV-KEY-001

Coordinate	EPSG:4548
Area Tolerance	±0.0% aligned
Scope	Conceptual design
Condition	Statutory permits

Beijing AI Origin Community (104.3 ha)

BEIJING AI ORIGIN COMMUNITY

Positioning	Origin dispatch & inclusive living district
Facilities	Physical dispatch window, night school, greenway
Governance	L0/L1 baseline; staffed-window floor (design target)
Boundary	Qinghuayuan heritage setback CONS-001

TECHNICAL DISCLOSURE • PROV-KEY-002

Coordinate	EPSG:4548
Area Tolerance	±0.0% aligned
Scope	Conceptual design
Condition	Statutory permits

Dazhongsi AI Industry Cluster (72.0 ha)

DAZHONGSI AI INDUSTRY CLUSTER

Positioning	Marshalling yard & commercialisation hub
Facilities	Heritage civic plaza, 4-station service desk
Governance	SEB v0.5.0 four-station sequence self-check (participant-declared)
Boundary	Dazhongsi heritage control zone CONS-002

TECHNICAL DISCLOSURE • PROV-KEY-003

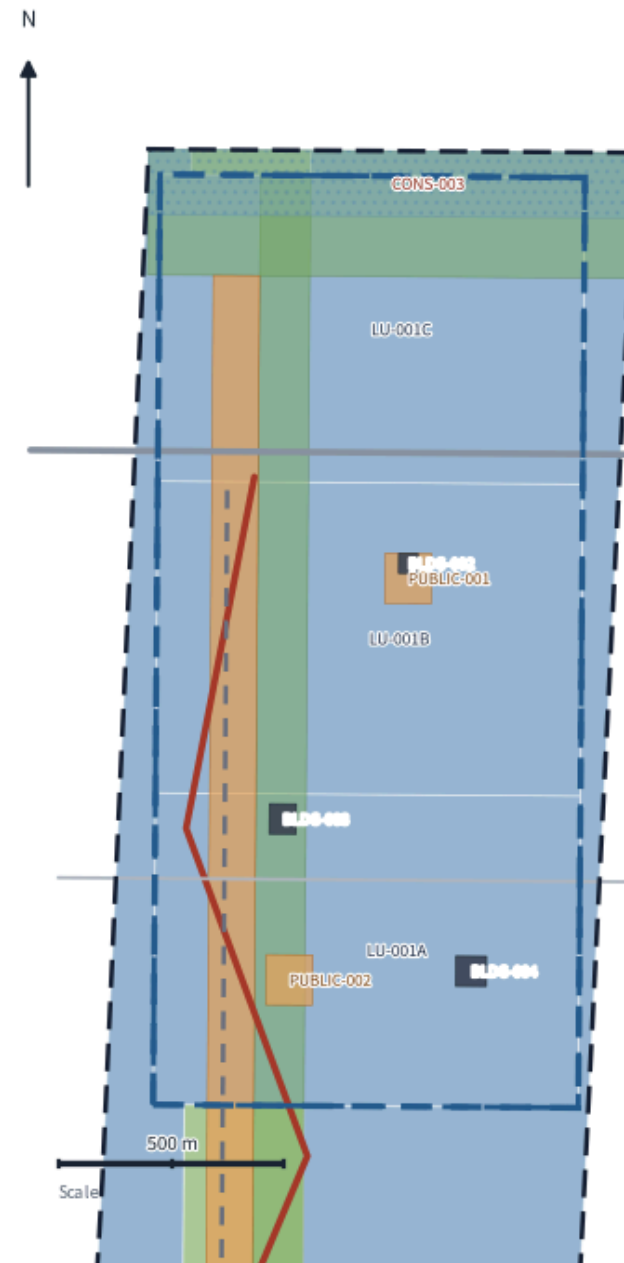
Coordinate	EPSG:4548
Area Tolerance	±0.0% aligned
Scope	Conceptual design
Condition	Statutory permits

面积为公告口径；几何为 provisional 粗略矩形，非地块红线；平面图见 key-area-plan-*.png（含 feature_id 标注）

ZHONGZHIYUAN AI ACCELERATION AREA PLAN (Test Track • provisional)

PROVISIONAL BOUNDARY (NON-STATUTORY)

Land-use bands • building footprints • public nodes • heritage setbacks • feature_id labels (viewport = key area + 350 m)



Legend (layers & feature_id)

- Land use LU-001A~LU-005 (colored by code)
- Green space GREEN-001~006
- Public space PUBLIC-000~012
- Building footprints BLDG-001~010
- Road centerlines ROAD-001~014 (greenway/arterial/expressway)
- Constraints CONS-001~004 (heritage/blue line/rail, indicative)
- Site boundary SITE-001 (provisional)

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BEIJING AI ORIGIN COMMUNITY PLAN (Departure Yard • provisional)

PROVISIONAL BOUNDARY (NON-STATUTORY)

Land-use bands • building footprints • public nodes • heritage setbacks • feature_id labels (viewport = key area + 350 m)



Legend (layers & feature_id)

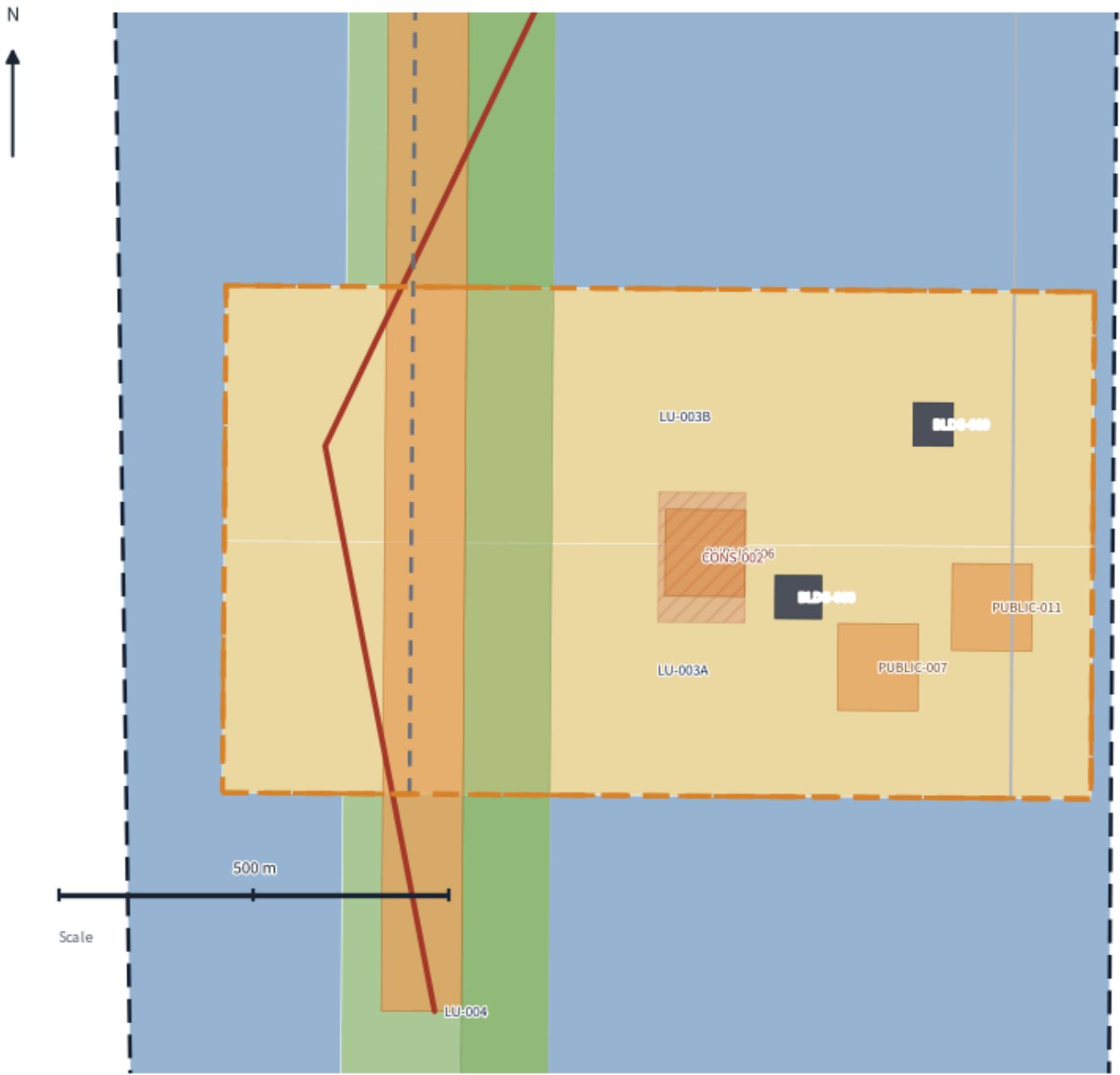
- Land use LU-001A~LU-005 (colored by code)
- Green space GREEN-001~006
- Public space PUBLIC-000~012
- Building footprints BLDG-001~010
- Road centerlines ROAD-001~014 (greenway/arterial/expressway)
- Constraints CONS-001~004 (heritage/blue line/rail, indicative)
- Site boundary SITE-001 (provisional)

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DAZHONGSI AI INDUSTRY CLUSTER PLAN (Marshalling Yard • provisional)

PROVISIONAL BOUNDARY (NON-STATUTORY)

Land-use bands • building footprints • public nodes • heritage setbacks • feature_id labels (viewport = key area + 350 m)



Legend (layers & feature_id)

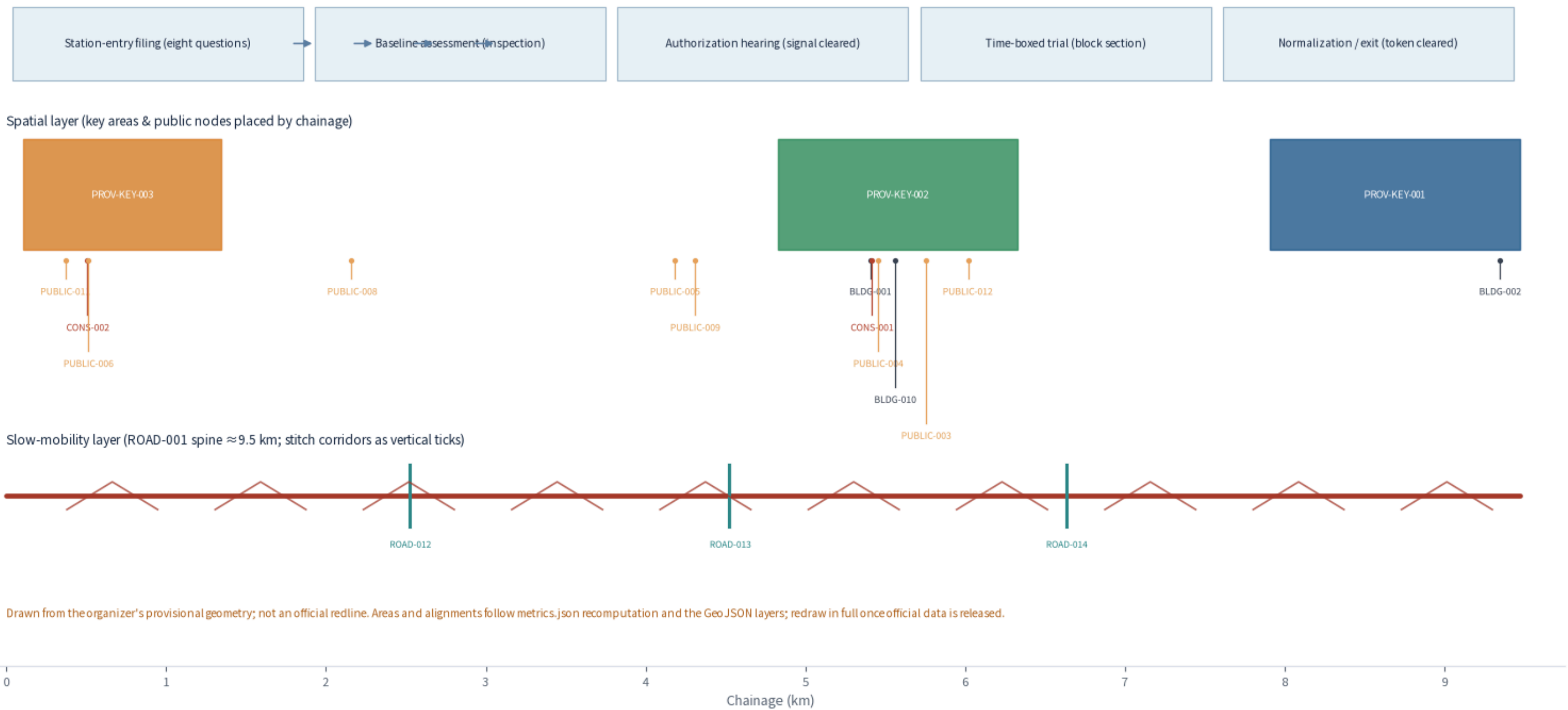
- Land use LU-001A~LU-005 (colored by code)
- Green space GREEN-001~006
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- Building footprints BLDG-001~010
- Road centerlines ROAD-001~014 (greenway/arterial/expressway)
- Constraints CONS-001~004 (heritage/blue line/rail, indicative)
- Site boundary SITE-001 (provisional)

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REPRESENTATIVE PROCESS SECTION OF THE SWITCHBACK SPINE (ROAD-001 chainage)

PROVISIONAL BOUNDARY (NON-STATUTORY)

Top: governance flow (four systems) • Middle: spatial layer (key areas & nodes) • Bottom: slow-mobility layer (spine & stitch corridors) • conceptual schematic, not a surveyed section



SLOW MOBILITY, TRANSIT & BLUE-GREEN PUBLIC SPACE NETWORK

PROVISIONAL GEOMETRY • EPSG:4548

≈9.5 km spine + stitch corridors ≈ 11.0 km slow network (recomputed) • Pedestrian Priority • The Human Switch (Tiered Safety Interface)

1. MULTI-TIERED MOBILITY NETWORK

Spine Continuous Greenway

≈9.5 km linear spine (recomputed) linking Beijing North to Tsinghua

AI Micro-Shuttle Test Lane

Speed cap < 15 km/h, dedicated lane, pedestrian priority

Barrier-Free & Silver Walkways

Full tactile paving & gentle slopes (design-floor target), emergency call points

Ring-Road Stitching Bridges

Bridging Ring 2/3/4 barriers for uninterrupted mobility

2. BLUE-GREEN SYSTEM & THE HUMAN SWITCH

Xiaoyuehe Sponge Green Infrastructure

178.94 ha green space, stormwater retention, microclimate

Origin Civic Plazas & Open Space

106.23 ha public space for community life and events

The Human Switch — Tiered Safety Interface

Public alert logged & displayed -> authorised on-site e-stop (long press + two-person confirm) -> professional recovery at dispatch hall; fully audited with accountability chain (conceptual; parameters are design assumptions pending field testing)

SEB v0.5.0 Hardwired Self-Check

Two-part handover ledger, staffed windows; 7/7 tabletop fixtures match (participant-declared, synthetic fixtures, no third-party certification)

QUANTITATIVE METRICS, SIMULATION LEDGER & EVIDENCE CROSSWALK

PROVISIONAL GEOMETRY • EPSG:4548

Spatial Recalculation • 10 Synthetic Tasks (90% Success) • SEB v0.5.0 Tabletop Self-Check (7/7 Match) • 7-Dimension Rubric

1. SPATIAL & SIMULATION METRICS

Metric Key	Declared Value	Provenance / Formula
Total Site Area (site_area_sqm)	11,412,825.39 m ²	EPSG:4548 provisional boundary
Green Space Area (green_space_area_sqm)	1,789,421.32 m ²	unary_union intersection
Green Ratio (green_ratio)	0.15679 (15.68%)	Exact ratio (topology-consistent)
Public Space Area (public_space_area_sqm)	1,062,301.12 m ²	unary_union intersection
Public Space Ratio (public_space_ratio)	0.09308 (9.31%)	Exact ratio (topology-consistent)
Simulation Task Count	10 tasks (SIM-001-010)	Deterministic seed: 20260831
Simulation Success Rate	0.90 (90.0%)	Honest disclosure of overrun
SEB Tabletop Self-Check	7 / 7 fixtures match (Exit 0)	Participant-declared • synthetic fixtures • no third-party certification

指标与 simulation.json / metrics.json 机器可读值一致；SEB 推演为参与者桌面自检，可离线复跑，未经第三方认证

2. RUBRIC CROSSWALK & DATA GAP DISCLOSURE

Brief Alignment (20%)

3-level scopes • 6 required tasks • 3 key areas • ≈11.0 km slow network (recomputed)

Originality (10%)

Working Diagram concept • Switchback unit • 4 innovation epochs

AI Innovation (15%)

4-system interlock • 12 scenario cards • SEB v0.5.0 self-check 7/7

Feasibility (20%)

JZ-01~10 lifecycle ledger • Adopter rights • 4-phase rollout

Public Interest (10%)

Staffed-window floor (design target) • Barrier-free • The Human Switch (tiered)

Risk Compliance (10%)

Provisional boundaries • Heritage setbacks • Adverse readings

Completeness (15%)

Bilingual parity • Embedded fonts • A3/A0 drawings • 4 Gate PASS

Statutory Notice: Official redlines & FAR are unknown; conceptual design does not replace statutory permits.